

EXACT RECONSTRUCTION OF REFLECTED-CAP DATA FROM AN ERASED FLAT BOUNDARY

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ABSTRACT. Let a convex three-polytope in a hyperplane of $\mathbb{R}^4$ carry one reflected pyramidal cap on each facet, with the caps independently rotated about their facets. We study the proper-zero regime: at least one cap remains nonflat and at least one cap returns to the core hyperplane. The flat caps then merge with the core and erase their internal source-layer boundaries.

We prove that every such three-dimensional proper-zero raw union has exactly one compatible decomposition. Starting from any recovered visible hinge, a frontier ridge exposes one new oriented wall ray. Its common ridge projection and distance from the recovered centre determine the reflected site; the unique strict-bisector component containing that site recovers the complete flat cap and its base. Connectivity of the facet-adjacency graph propagates the construction to every hidden facet. No genericity or bound on the number of facets is required.

For each fixed realized core and visible-facet mask, an auxiliary intrinsic boundary-germ skeleton gives a one-shot signature reconstruction away from a finite union of proper spheres and quadrics in the centre. We retain two independent structural consequences for hypothetical competing decompositions: differing sites avoid visible hinges and must number at least five on each side. The latter rests on a nonexistence theorem for an eight-point spherical configuration, proved by four polynomial certificate identities and an elementary positivity analysis.

We give exact rational implementations of the signature kernel and of the global propagation kernel. The independent audit completes 61,030 global propagation/component steps over 16,759 exact data with no mismatch, including exceptional centres and erased straight ridges; four deliberate orientation, halfspace, component and ridge-owner faults are rejected. Finite replay supports the implementation and is not used as the proof.

1. INTRODUCTION

Inverse Voronoi and Dirichlet problems usually begin with a supplied tessellation, cell complex, embedded diagram, or collection of cell boundaries. The input considered here is coarser. Several reflected sites and their facet boundaries have been erased by a coplanar union before the inverse problem begins.

The model is elementary. A convex polytope P lies in a three-dimensional hyperplane $H \subset \mathbb{R}^4$, an interior point o is fixed, and the reflection of o in each facet plane is used as the apex of a pyramid. Each pyramid is then

rotated through an independent angle about its base facet. When an angle is zero, the corresponding pyramid is coplanar with P . All such flat pyramids merge with P into one three-dimensional subset $M_J \subset H$. Only the raw union is observed; cell labels and internal flat seams are not.

The central issue is therefore not the forward reflection construction. That construction is classical and appears explicitly in Ash and Bolker's Theorem 2 [3]. The issue is whether the hidden reflected sites can be identified from the exposed boundary after the internal seams have disappeared.

A visible hinge supplies more than an initial site estimate: it orients one side of every frontier ridge. In a normal two-dimensional section the bare boundary then has exactly one other oriented ray, even when the two rays form one straight line after a coplanar merger. This observation supports a finite propagation through the facet-adjacency graph and removes the exceptional-centre hypothesis from uniqueness.

Informal theorem. *Every three-dimensional proper-zero reflected-cap raw union determines its core, centre, hidden reflected sites, flat facets and caps uniquely. Hence congruent raw unions arise from congruent data. No genericity hypothesis and no bound on the number of hidden facets is required.*

A second, one-shot reconstruction uses the noncoplanar crease graph of maximal exposed planar sheet germs. It keeps a collinear point exactly when its local incident-sheet germ changes and removes a pure presentation subdivision. If no core vertex passes its perpendicular-bisector neighbour test, the passing vertices are precisely the hidden reflected sites. For a fixed realized core and visible mask, the centres at which this shortcut can acquire a false candidate lie in a measure-zero algebraic set.

Informal theorem. *Under a hypothetical pair of distinct compatible decompositions, every differing site avoids all visible hinges and the two site sets differ by at least five elements on each side. These implications provide independent structural checks, although the universal theorem rules out the antecedent in the present model.*

The five-site bound reduces, through an exchange graph and an elementary incidence count, to the nonexistence of a specific eight-point spherical configuration. That nonexistence is proved by a computer-assisted but fully verifiable argument: four explicit polynomial identities, found by a Gröbner search and re-verified independently by exact expansion, force the configuration into a one-parameter family that an elementary positivity analysis then empties. Section 7 contains the complete argument.

The theorem is specific to the proper-zero regime: at least one visible nonflat cap is needed to recover the initial hinge and centre. We do not claim all-zero uniqueness, a result in higher dimension, stability under noise, or novelty or priority relative to all possible prior art. Those boundaries are repeated in Section 10.

2. THE REFLECTED-CAP MODEL AND EXACT CONTACTS

Let $H \cong \mathbb{R}^3$ be a fixed hyperplane in $\mathbb{R}^4$, and let e be a unit normal to H . Let $P \subset H$ be a full-dimensional convex polytope and $o \in \text{int } P$. Translate o to the origin and write the irredundant unit-normal description

$$P = \{x \in H : n_i \cdot x \leq h_i, \ i = 1, \dots, m\}, \quad \|n_i\| = 1, \quad h_i > 0,$$

with facets $F_i = P \cap \{n_i \cdot x = h_i\}$. Write

$$H_i^- = \{x \in H : n_i \cdot x \leq h_i\}$$

for the inward closed halfspace of the i th facet. For $\theta_i \in (-\pi, \pi)$ set

$$a_i(\theta_i) = h_i(1 + \cos \theta_i)n_i + h_i \sin \theta_i e, \quad C_i(\theta_i) = \text{conv}(F_i \cup \{a_i(\theta_i)\}).$$

The excluded value π collapses the apex to o . At $\theta_i = 0$, $a_i = 2h_i n_i$, the reflection of o in $\text{aff } F_i$: at zero angle the sites are exactly those of Ash and Bolker's forward construction [3].

Two data are called congruent when an ambient Euclidean isometry carries the core, centre, and unordered cap family of one to those of the other. Thus the angle coordinates are used only to parametrise the geometric caps.

The datum is $D = (P, o, \theta)$ and its raw union is

$$X(D) = P \cup \bigcup_{i=1}^m C_i(\theta_i) \subset \mathbb{R}^4,$$

retained as a bare Euclidean subset. Put

$$J = \{i : \theta_i = 0\}, \quad I = [m] \setminus J.$$

The datum is *proper-zero* when I and J are both nonempty. Its merged flat component is

$$M_J = P \cup \bigcup_{j \in J} C_j(0) \subset H.$$

Proposition 2.1 (exact contacts). *For every angle vector $\theta \in (-\pi, \pi)^m$,*

$$P \cap C_i(\theta_i) = F_i, \quad C_i(\theta_i) \cap C_j(\theta_j) = F_i \cap F_j \quad (i \neq j).$$

Thus the cells form an embedded face-to-face polyhedral complex whose contact poset is independent of the angles.

Proof. Use the Voronoi diagram of the sites $0, a_1(\theta_1), \dots, a_m(\theta_m)$. The sites are distinct: equality of two nonzero H -components would force two irredundant outward facet normals to agree. For $x \in P$,

$$\begin{aligned} \|x - a_i\|^2 - \|x\|^2 &= \|a_i\|^2 - 2x \cdot a_i \\ &= 2h_i(1 + \cos \theta_i)(h_i - n_i \cdot x) \geq 0. \end{aligned}$$

The first factor is positive on $(-\pi, \pi)$. Hence P lies in the Voronoi cell V_0 of 0 , and $F_i \subset V_0 \cap V_i$. The apex a_i is in the interior of V_i . Convexity of V_i gives $C_i \setminus F_i \subset \text{int } V_i$. Therefore no point of $P \cap C_i$ can lie outside F_i .

If a point of $C_i \cap C_j$ lay outside one of the two hinge facets, the same argument would place it in one open Voronoi cell and simultaneously in

a different open cell or in V_0 . This is impossible. The reverse inclusion $F_i \cap F_j \subset C_i \cap C_j$ is immediate. $\square$

3. INTRINSIC RECOVERY IN THE PROPER-ZERO REGIME

For a three-dimensional polyhedral subset $Y \subset \mathbb{R}^4$, let $\text{FlatReg}_3(Y)$ be the set of points having a neighbourhood in Y that is relatively open in one affine three-plane. This set is determined by Y alone.

Proposition 3.1 (proper-zero intrinsic recovery). *If $I \neq \emptyset$, the raw subset $X(D)$ intrinsically determines M_J , every nonflat cap and its hinge facet, the centre o , and the fully reflected site associated with every nonflat hinge.*

Proof. The closures of the connected components of $\text{FlatReg}_3(X)$ are M_J and the nonflat caps. Indeed, the zero-angle cells join the core through facets inside H and form one three-dimensional flat component, while every nonflat cap has a different affine hull. Proposition 2.1 accounts for every contact. The codimension-one adjacency graph of these closures is a star.

If $|I| \geq 2$, the unique star centre is M_J . If $|I| = 1$, let i be the nonflat index, let $V = \text{vol}_3(P)$, and let $v_k = \text{vol}_3 \text{conv}(\{o\} \cup F_k)$. The radial pyramids tile P , so $\sum_k v_k = V$. Rotation preserves the intrinsic volume of a cap, and hence

$$\text{vol}_3(M_J) = 2V - v_i > v_i = \text{vol}_3(C_i).$$

Thus in the two-component case the larger component is M_J .

Fix one recovered nonflat cap, with apex A_i and hinge F_i . Let p_i be the orthogonal projection of A_i to $\text{aff } F_i$, let $r_i = \text{dist}(A_i, \text{aff } F_i)$, and let ν_i be either unit normal to $\text{aff } F_i$ inside H . Rotation about $\text{aff } F_i$ preserves the radius in its two-dimensional normal plane, so

$$\{p_i - r_i \nu_i, p_i + r_i \nu_i\} = \{o, a_i(0)\}.$$

The centre belongs to the interior of M_J . The other point does not belong to M_J : apply Proposition 2.1 to the angle vector in which cap i is also set to zero. Its apex meets neither the core nor another cap away from prescribed lower-dimensional contacts. Consequently o is the unique one of the two candidates in $\text{int } M_J$. Reflection of o in the recovered hinge then gives the fully reflected site. $\square$

Every compatible decomposition of a given proper-zero raw union therefore has the same M_J , centre, nonflat caps, and visible hinge facets.

4. UNIVERSAL RIDGE PROPAGATION AND THE INTRINSIC BOUNDARY-GERM SKELETON

Definition 4.1 (compatible decomposition). A compatible decomposition of M_J consists of a full-dimensional convex core P' with $o \in \text{int } P'$ and hidden facets G_a indexed by sites a , where the facet set of P' is exactly the already

recovered visible hinge facets together with the pairwise distinct facets G_a , each site a is the reflection of the recovered centre o in $\text{aff } G_a$, and

$$M_J = P' \cup \bigcup_{a \in A} \text{conv}(G_a \cup \{a\})$$

with pairwise disjoint cell interiors. The original datum is one such decomposition. For an edge f of G_a , write $W(a, f) = \text{conv}(\{a\} \cup f)$ for the corresponding wall.

In what follows, a *proper-zero raw union* means $X(D)$ for a proper-zero datum D . A compatible decomposition of $X(D)$ means a compatible decomposition of its intrinsically recovered M_J , together with the nonflat caps recovered by Proposition 3.1. Thus the phrases used for a datum, its raw union, and its merged flat component do not denote different inverse problems.

Lemma 4.2 (boundary inventory). *For every compatible decomposition,*

$$\partial M_J = \bigcup_{i \in I} F_i \cup \bigcup_{a \in A} \bigcup_{f \in E(G_a)} W(a, f).$$

Proof. The boundary of P' is the union of its visible facets and hidden facets. Each hidden facet is shared with its cap and becomes internal to M_J . The remaining cap faces are the walls. Proposition 2.1 says that a wall meets another cell only in a lower-dimensional face. $\square$

4.1. Universal propagation from a visible hinge. The next argument works directly with the oriented local boundary of M_J . It does not classify a point by the size of an angle and does not use the auxiliary intrinsic germ skeleton of Subsection 4.2.

Lemma 4.3 (frontier-ridge exposure). *Suppose a facet F of a compatible core, and the flat cap over F when it exists, have already been recovered. Let R be an edge of F whose other incident core facet G is hidden and has not yet been recovered. For $x \in \text{relint } R$, the local boundary of M_J in the oriented two-dimensional normal quotient to R determines the transverse wall ray of the cap over G . This remains true when that ray and the known boundary ray are opposite rays of one line.*

Proof. Proposition 2.1 excludes a third cell from a neighbourhood of x away from the common ridge. The quotient origin is the image of R (equivalently of the recovered point x), so it is intrinsic and splits even a straight boundary line canonically into two oriented rays. Orient the ray of F at angle 0, put the ray of G at angle $\alpha \in (0, \pi)$, and write the direction of the projected centre as $\beta \in (0, \alpha)$. Reflection in the line of G puts its flat-cap ray at angle $2\alpha - \beta$.

If F is visible, the local union occupies the angular interval $[0, 2\alpha - \beta]$; its known boundary ray is the ray at 0. If F is hidden, its already recovered cap extends the interval to $[-\beta, 2\alpha - \beta]$; its known boundary ray is the ray

at $-\beta$. Thus the other oriented boundary ray is always the transverse wall ray of the cap over G . It cannot have the same orientation as the known ray: $2\alpha - \beta = 0$ is incompatible with $0 < \beta < \alpha$ in the visible case, and $2\alpha - \beta = -\beta$ would give $\alpha = 0$ in the hidden case. The two rays may be opposite, so their supporting lines may merge into one straight line; retaining ray orientation still distinguishes them. No erased source seam is used. $\square$

Lemma 4.4 (site recovery from a ridge ray). *In the setting of Lemma 4.3, let q be the orthogonal projection of o onto $\text{aff } R$, let $r = \text{dist}(o, \text{aff } R)$, and let u be the unit transverse direction of the newly exposed wall ray. Then the reflected site over G is*

$$a_G = q + ru.$$

Proof. Both reflection in $\text{aff } G$ and orthogonal projection to the quotient normal to $\text{aff } R$ fix R . Hence o and its reflection a_G have the same projection q on $\text{aff } R$ and the same distance r from it. The wall $\text{conv}(R \cup \{a_G\})$ supplies the oriented quotient direction u , which selects the displayed point rather than its antipode. $\square$

Lemma 4.5 (site-anchored cap component). *Let a be a recovered hidden site and put*

$$B_a = \{y \in H : \|y - a\| = \|y - o\|\}, \quad H_a^+ = \{y \in H : \|y - a\| < \|y - o\|\}.$$

Then the unique connected component of $M_J \cap H_a^+$ containing a is $C_a \setminus G_a$. Its closure is the entire flat cap C_a , and

$$G_a = \overline{\text{Comp}_a(M_J \cap H_a^+)} \cap B_a.$$

Consequently the raw set, o , and a determine both the cap and its base.

Proof. The bisector B_a is $\text{aff } G_a$. The core lies in the closed halfspace on the o -side, whereas every point of $C_a \setminus G_a$ lies strictly on the a -side. Therefore $C_a \cap H_a^+ = C_a \setminus G_a$, which is connected and contains a . By Proposition 2.1, every contact of C_a with another cell is contained in $G_a \subset B_a$. Inside the strict halfspace H_a^+ its piece is consequently disjoint from all other cell pieces. Because the cell family is finite and each cell is closed, this piece and its complement are both closed relative to $M_J \cap H_a^+$; hence it is a connected component. No other component contains a . Every point of the base is approached along its segment to a , so the closure of the component is C_a , and intersecting that closure with the bisector recovers exactly G_a . $\square$

Theorem 4.6 (exact reconstruction). *Every three-dimensional proper-zero datum has exactly one compatible decomposition. More precisely, its raw union determines M_J , the centre, the core, every reflected site, and every flat and nonflat cap. Consequently, $X(D)$ congruent to $X(D')$ implies D congruent to D' .*

Proof. Proposition 3.1 supplies M_J , o , all nonflat caps, and their visible hinge facets. At least one such facet exists. Begin with all visible facets marked as

recovered. Maintain the invariant that a marked hidden facet comes with its complete recovered cap, hence with the known wall ray at every one of its edges. The facet-adjacency graph of a three-polytope is connected. If not every facet has been recovered, it has a frontier edge R belonging to a recovered facet F and to an unrecovered hidden facet G . Lemma 4.3 obtains the oriented new wall ray from the bare local boundary, Lemma 4.4 recovers the site, and Lemma 4.5 recovers the full cap and its base G , preserving the invariant. Marking G recovered and repeating terminates after finitely many facets.

To see that this construction is independent of a chosen compatible decomposition, compare any two of them. Their initial visible facets and centre agree by Proposition 3.1. At a frontier edge of the first decomposition, the other facet of the second cannot already be a different recovered facet. Indeed, recovered polygons are common to both decompositions; such a facet would also contain R in the first decomposition, producing a third facet incident to one polyhedral edge, which is impossible. The same intrinsic oriented ray therefore gives the same site, cap, and base in both decompositions. Induction through the connected facet-adjacency graph makes every facet, site, and cap of the first decomposition a facet, site, and cap of the second.

The recovered facet polygons form the complete boundary of the first convex core. The second core contains every one of their vertices and lies in every corresponding inward halfspace, so convexity gives both inclusions and the cores coincide. Its irredundant facet set admits no additional facet. Thus the two decompositions are equal. Finally, an ambient isometry carrying one raw union to another transports the intrinsic construction and hence the complete datum. $\square$

4.2. The auxiliary intrinsic germ skeleton. Let an *exposed planar sheet* of ∂M_J be the closure of a maximal connected planar piece of its relative-regular locus. Internal coplanar subdivisions are not retained.

Definition 4.7 (intrinsic germ skeleton). Atomise the boundaries of all exposed sheets at every endpoint contributed by any other sheet. Retain the open segments carried by at least two distinct supporting planes. Suppress a collinear degree-two point only if its complete local incident-sheet germ is unchanged through that point. The resulting canonical graph is $\Gamma(M_J)$. A vertex a *passes* if every neighbour of a in $\Gamma(M_J)$ belongs to the perpendicular bisector

$$B(o, a) = \{x \in H : \|x - o\| = \|x - a\|\}.$$

The definition is intrinsic: exposed sheet subsets, their supporting planes, and their local germs are properties of M_J as a point set. In particular, the graph stores neither a hidden cap label nor an erased internal seam.

Lemma 4.8 (decomposition computation of the intrinsic graph). *Starting from the inventory in Lemma 4.2, erase a side across which the two incident boundary patches are coplanar, atomise at every other sheet-boundary*

endpoint, and suppress only unchanged collinear degree-two germs. The result is $\Gamma(M_J)$, independently of the compatible decomposition used for the computation.

Proof. At a relative-interior point of a patch side, exact face-to-face contact gives no transverse crossing in a patch interior. If the two incident patch planes agree, their union is locally planar and the side is absent from the crease locus. If the planes differ, the side is an intrinsic noncoplanar crease. Passing to maximal planar sheets deletes exactly the first kind. Atomisation restores every endpoint where another sheet germ begins, ends, or changes, including a collinear endpoint omitted by a polygon hull. The final suppression removes only a presentation subdivision at which the complete local germ is unchanged. These conditions depend only on ∂M_J . $\square$

Remark 4.9 (the swallowed-corner control). There are legitimate data for which two coplanar merged triangles have a collinear boundary point that a polygonal hull would drop. At that point a noncoplanar incident sheet changes, so Definition 4.7 retains it. The exact control used in the artifact is

$$P = \{x \geq 0, z \geq 0, x + y \geq 2, y \leq 6, x + z \leq 4\}, \quad o = (1, 2, 1),$$

with the hidden facets $x = 0$ and $z = 0$. Their shared edge starts at the foot of the perpendicular from o , making the two apexes and that endpoint collinear.

5. THE APEX SIGNATURE AND UNIQUENESS

Lemma 5.1 (only its own walls contain an apex). *For $a \in A$, the boundary patches containing a are exactly the walls of the cap with apex a .*

Proof. The reflection a lies strictly outside P' . For $b \neq a$, Proposition 2.1 gives $C_a \cap C_b = G_a \cap G_b \subset P'$, so a belongs to no wall of C_b . Lemma 4.2 leaves no other boundary patch. $\square$

Lemma 5.2 (candidate types). *Every vertex of $\Gamma(M_J)$ is either a hidden apex or a vertex of the compatible core.*

Proof. By Lemma 4.8, a graph vertex is a branch point or a retained endpoint of the patch-side arrangement. Exact face-to-face contacts create no transverse crossing in a patch interior, so such a point is a patch corner. Visible-facet corners are core vertices, and wall corners are one apex and two core vertices. $\square$

Lemma 5.3 (apexes pass). *Every hidden apex passes in $\Gamma(M_J)$.*

Proof. Let a be the apex over a hidden facet G_a . For each vertex v of G_a , the two walls over the two facet edges incident to v meet along $[a, v]$. Their planes are distinct: if they were coplanar, the apex and three consecutive vertices of the convex facet polygon would be coplanar, forcing $a \in \text{aff } G_a$, contrary to $o \in \text{int } P'$. Thus the relative interior of $[a, v]$ is an intrinsic crease. The apex is a branch point, and v is retained because the two wall-sheet

germs containing a end or change there. Conversely, Lemma 5.1 shows that every crease incident to a is of this form. Hence

$$N_{\Gamma(M_J)}(a) = \text{vert } G_a \subset \text{aff } G_a = B(o, a).$$

□

For a facet f of a compatible core, write $v_f = \text{vol}_3 \text{conv}(\{o\} \cup f)$.

Lemma 5.4 (volume identity). *Let $V_{\text{vis}} = \sum_{i \in I} v_{F_i}$. Every compatible decomposition satisfies*

$$\text{vol}_3(M_J) = V_{\text{vis}} + 2 \sum_{a \in A} v_{G_a}.$$

Consequently all compatible cores have the same volume.

Proof. The radial pyramids from o to the facets tile a convex core, so $\text{vol}_3(P') = V_{\text{vis}} + \sum_a v_{G_a}$. The cap over G_a is a reflection of its radial pyramid and has the same volume. Exact contacts give pairwise disjoint cell interiors, proving the displayed identity. Both M_J and V_{vis} are intrinsic, so $\sum_a v_{G_a}$ and then the core volume are common to all decompositions. □

Lemma 5.5 (nested site sets). *If (P', A) and (P'', A'') are compatible and $A'' \subseteq A$, then the two decompositions coincide.*

Proof. The hidden facet associated with a has halfspace $\{x : \|x - o\| \leq \|x - a\|\}$, the Dirichlet-cell description of the core against its reflected sites [3]. Therefore

$$P' = \bigcap_{i \in I} H_i^- \cap \bigcap_{a \in A} \{x : \|x - o\| \leq \|x - a\|\},$$

and similarly for P'' . The inclusion $A'' \subseteq A$ gives $P' \subseteq P''$. Lemma 5.4 gives equal finite volume, so the two closed convex bodies agree. Their active facets and reflected sites then agree. □

Theorem 5.6 (signature reconstruction). *Let D be a three-dimensional proper-zero datum. Suppose no vertex of its core passes in $\Gamma(M_J)$. Then the passing vertices of $\Gamma(M_J)$ are exactly the hidden reflected apexes, the core is*

$$P = \bigcap_{i \in I} H_i^- \cap \bigcap_{a \text{ passing}} \{x : \|x - o\| \leq \|x - a\|\},$$

and every compatible decomposition of $X(D)$ is the same. Consequently, $X(D)$ congruent to $X(D')$ implies D congruent to D' .

Proof. Let A be the hidden apex set of D , and let (P'', A'') be any compatible decomposition. The graph $\Gamma(M_J)$ is intrinsic, hence common to both decompositions. Lemma 5.3 gives that every element of A and A'' passes in this common graph. By Lemma 5.2, a passing vertex is an apex of D or a core vertex. The second possibility is excluded by hypothesis, so the passing set is exactly A and $A'' \subseteq A$. Lemma 5.5 completes the uniqueness proof.

An ambient isometry carrying one raw union to another preserves the intrinsic components, centre, exposed sheet germs, and $\Gamma(M_J)$. It carries the first datum to a compatible decomposition of the second raw union, which uniqueness identifies with the second datum. $\square$

6. THE EXCEPTIONAL CENTRE SET FOR THE SIGNATURE SHORTCUT

Theorem 6.1 (fixed-realization genericity). *Fix a realized three-polytope P and a visible-facet mask. The set of centres $o \in \text{int } P$ at which some core vertex passes is contained in a finite union of proper spheres and quadrics. In particular, it has three-dimensional Lebesgue measure zero.*

Proof. Use fixed absolute coordinates. Suppose a core vertex v is incident to a hidden facet with outward unit normal n and equation $n \cdot x = h$. The reflected apex is

$$a = o + 2(h - n \cdot o)n.$$

The two walls meeting along $[v, a]$ have distinct planes, so this is an edge of $\Gamma(M_J)$. If v passes, then $a \in B(o, v)$, which forces

$$4(h - n \cdot o)^2 = \|v - o\|^2.$$

Its quadratic part is $o^\top(4nn^\top - I)o$. The eigenvalue along n is 3 and the two eigenvalues on $n^\perp$ are -1 . The polynomial is nonzero and its zero set is a proper quadric.

If v is incident only to visible hinge facets, the core edges at v are intersections of distinct facet planes and are intrinsic crease edges. Any crease neighbour w required by passing satisfies

$$\|w - o\|^2 = \|w - v\|^2,$$

a proper sphere condition in o .

There are finitely many vertices and incidences. Every passing event is contained in at least one of the displayed proper algebraic hypersurfaces, and a finite union of such zero sets has measure zero. $\square$

This exceptional set limits only the one-shot signature shortcut of Theorem 5.6; it is not an exceptional set for uniqueness, which is unconditional in the proper-zero regime by Theorem 4.6. The theorem deliberately does not assert that the bad-centre set itself is closed. A subset of a closed algebraic union need not be closed. Nor does the argument define a measure on simultaneously varying realizations of P .

7. STRUCTURE OF A HYPOTHETICAL COMPETING DECOMPOSITION

Theorem 4.6 rules out a failure of uniqueness in the present proper-zero model. This section retains an independent conditional route: it studies the structure that a hypothetical pair of compatible decompositions would have and supplies checks that do not use ridge propagation. Throughout, D is a three-dimensional proper-zero datum with decomposition (P, A) and centre o , and (P'', A'') is a second compatible decomposition of the same raw union.

By Section 3 the two decompositions share M_J , the centre, the visible hinge facets, and the intrinsic graph $\Gamma(M_J)$.

Lemma 7.1 (differing sites avoid visible hinges). *Let $a \in A'' \setminus A$. Then a is a vertex of P that passes in $\Gamma(M_J)$ and lies on no visible hinge facet. Symmetrically for $A \setminus A''$.*

Proof. As an apex of the second decomposition, a passes (Lemma 5.3) and, by Lemma 5.2 applied to the first decomposition, a is an apex of D or a vertex of P ; the first option is excluded by $a \notin A$. Applying Lemma 5.1 in the second decomposition, the only boundary patches containing a are the walls of its own cap there. A visible hinge facet F_i is a boundary patch of every decomposition and is contained in the core of each; since an apex lies strictly outside the core, F_i is not a wall of the cap of a , and hence $a \notin F_i$. $\square$

Theorem 7.2 (hinge-anchored uniqueness). *Suppose every vertex of the core that passes in $\Gamma(M_J)$ lies on at least one visible hinge facet. Then the compatible decomposition is unique. Theorem 5.6 is the special case in which no core vertex passes. This criterion is independent of the universal propagation proof in Theorem 4.6.*

Proof. Suppose $(P'', A'') \neq (P, A)$. If $A'' \subseteq A$, Lemma 5.5 forces equality, so $A'' \setminus A \neq \emptyset$. Any $a \in A'' \setminus A$ is, by Lemma 7.1, a passing core vertex on no visible hinge facet, contradicting the hypothesis. $\square$

Lemma 7.3 (exchange edges). *Let $a \in A'' \setminus A$. Then a lies on at least three hidden facets of (P, A) . For every hidden facet $G_c \ni a$, its site c satisfies $c \in A \setminus A''$, the segment $[a, c]$ is an edge of $\Gamma(M_J)$, and the triangle (o, a, c) is equilateral. Symmetrically with the two decompositions exchanged.*

Proof. By Lemma 7.1, a is a vertex of the three-polytope P , so at least three facets of P pass through it, and none of them is visible; distinct facets have distinct planes, and the site is the reflection of o in the plane, so the corresponding sites are distinct.

Let G_c be such a hidden facet. Then $a \in \text{vert } G_c$, so by the proof of Lemma 5.3 the segment $[c, a]$ is an edge of $\Gamma(M_J)$. The facet plane $\text{aff } G_c$ is the perpendicular bisector of o and c , and a lies on it, so $\|a - o\| = \|a - c\|$. Since a passes and c is one of its graph neighbours, $\|c - o\| = \|c - a\|$. The triangle is equilateral.

Finally $c \notin A''$: otherwise c would be an apex of the second decomposition, and Lemma 5.3 applied there would give $N_{\Gamma(M_J)}(c) = \text{vert } G_c'' \subset P''$; but $a \in N_{\Gamma(M_J)}(c)$ is an apex of the second decomposition and lies strictly outside P'' . $\square$

Lemma 7.4 (two common neighbours). *Let $\mathcal{E}$ be the graph on $(A \setminus A'') \cup (A'' \setminus A)$ whose edges are the pairs $\{a, c\}$ of Lemma 7.3. Then $\mathcal{E}$ is simple and bipartite between the two difference sets, with minimum degree at least three, and two distinct sites on one side have at most two common neighbours.*

Proof. Bipartiteness, simplicity and the degree bound are Lemma 7.3. Let c be a common neighbour of the distinct same-side sites a and a' . The two equilateral relations give $\|c - o\| = \|c - a\| = \|a - o\|$ and $\|c - o\| = \|c - a'\| = \|a' - o\|$. So c lies on the perpendicular bisector plane of o and a , on that of o and a' , and on the sphere of radius $\|a - o\|$ about o . The two planes are distinct, because reflecting o across them returns a , respectively a' ; hence they meet in a line or not at all, and a line meets a sphere in at most two points. $\square$

Lemma 7.5 (incidence count). *Each side of $\mathcal{E}$ has at least four sites. If some side has exactly four, then both do, $\mathcal{E}$ is 3-regular, that is, $K_{4,4}$ minus a perfect matching, and all eight sites lie at one common distance $R > 0$ from o with every edge a chord of length R .*

Proof. Write p and q for the two side sizes. Every vertex has at least three neighbours, all on the other side, so $p, q \geq 3$. Count the incidences $(\{a, a'\}, c)$ of a common neighbour c with an unordered pair of distinct sites on the p -side: Lemma 7.4 bounds them by $2\binom{p}{2}$, and they number $\sum_c \binom{\deg c}{2} \geq 3q$, because each of the q vertices c of the other side has degree at least three.

If $p = 3$, every vertex of the other side is adjacent to all three sites, so one pair on the p -side has $q \geq 3 > 2$ common neighbours, impossible. Hence, exchanging the roles of the sides, $p, q \geq 4$. If $p = 4$, then $3q \leq 2\binom{4}{2} = 12$ forces $q \leq 4$, hence $q = 4$; and a vertex of degree four on either side would give $\sum_c \binom{\deg c}{2} \geq \binom{4}{2} + 3\binom{3}{2} = 15 > 12$, so every degree is exactly three. The bipartite complement of a 3-regular bipartite graph on $4 + 4$ vertices is 1-regular, a perfect matching. Such a graph is connected (a component would need at least three vertices on each side), and the equilateral relation of Lemma 7.3 propagates the distance $\|a - o\|$ across every edge, so all eight sites lie at one distance $R > 0$ and every edge has length R . $\square$

The remaining case is a genuinely geometric one, and it is the heart of the bound.

Theorem 7.6 (no exchange quadruple). *There is no configuration of distinct unit vectors $d_1, \dots, d_4$ and distinct unit vectors $e_1, \dots, e_4$ in $\mathbb{R}^3$ with*

$$e_l \cdot d_i = \frac{1}{2} \quad \text{for all } l \neq i.$$

Proof. Write $x_{ij} = d_i \cdot d_j$, let G be the unit-diagonal Gram matrix of $d_1, \dots, d_4$, let G_i be its principal 3×3 minor deleting row and column i , and let r_i be the deleted off-diagonal column. Define the polynomials

$$\begin{aligned} g_0 &:= \det G, \\ g_i &:= \mathbf{1}^\top \operatorname{adj}(G_i) \mathbf{1} - 4 \det G_i \quad (i = 1, \dots, 4), \\ s_i &:= \mathbf{1}^\top \operatorname{adj}(G_i) r_i - \det G_i. \end{aligned}$$

For example, in the variables (x_{23}, x_{24}, x_{34}) ,

$$g_1 = 3x_{23}^2 + 3x_{24}^2 + 3x_{34}^2 - 8x_{23}x_{24}x_{34} \\ + 2(x_{23}x_{24} + x_{23}x_{34} + x_{24}x_{34}) - 2(x_{23} + x_{24} + x_{34}) - 1.$$

Step 1: the configuration satisfies the polynomial system. Four vectors in $\mathbb{R}^3$ give $g_0 = \det G = 0$. Fix l and suppose the triple $\{d_i : i \neq l\}$ were linearly dependent; applying $e_l \cdot (\cdot)$ to a dependence shows the coefficients sum to zero, so three distinct points would be affinely dependent, hence collinear, and a line meets the unit sphere in at most two points. So each triple is a basis and $\det G_l \neq 0$, and e_l is the unique solution of the three linear equations $e_l \cdot d_i = \frac{1}{2}$ ($i \neq l$). Solving that system by Cramer's rule and imposing $\|e_l\|^2 = 1$ yields $\mathbf{1}^\top \text{adj}(G_l) \mathbf{1} = 4 \det G_l$, that is, $g_l = 0$ on the Gram coordinates of the configuration. Moreover $c_l := e_l \cdot d_l - \frac{1}{2}$ satisfies $2c_l \det G_l = s_l$. If some $c_l = 0$, then $e_l \cdot d_i = \frac{1}{2}$ for all four i , so all four d_i lie on the plane $\{x : e_l \cdot x = \frac{1}{2}\}$, which misses the origin; any three of them are then linearly independent, and every e_m solves the same three linear equations as e_l , forcing $e_m = e_l$ for all m and contradicting distinctness. Hence every $s_l \neq 0$.

Step 2: four certificate identities force an isosceles pattern. There exist explicit polynomials $q_0^{(\ell)}, \dots, q_4^{(\ell)} \in \mathbb{Q}[x_{12}, \dots, x_{34}]$ of total degree at most eight with

$$\ell \cdot s_1 s_2 = q_0^{(\ell)} g_0 + q_1^{(\ell)} g_1 + q_2^{(\ell)} g_2 + q_3^{(\ell)} g_3 + q_4^{(\ell)} g_4$$

for each of the four linear forms

$$\ell \in \{x_{12} - x_{34}, x_{13} - x_{24}, x_{14} - x_{23}, x_{23} + x_{24} + x_{34} + 1\}.$$

The cofactors are included, with the verification script, in the companion artifact described in Section 8; each identity is re-verified by expanding the difference of the two sides to the zero polynomial in exact rational arithmetic, independently of the software that found it. At the configuration all g_i vanish and $s_1 s_2 \neq 0$, so all four linear forms vanish:

$$x_{12} = x_{34} =: a, \quad x_{13} = x_{24} =: b, \quad x_{14} = x_{23} =: c, \quad a + b + c = -1.$$

Step 3: an elementary emptiness argument. Substituting the pattern into any of the four face polynomials leaves one cubic; explicitly,

$$g_i|_{\text{pattern}} = 4F(a, b), \quad F(a, b) = 2ab^2 + 2a^2b + a^2 + 3ab + b^2 + a + b + 1,$$

an identity in (a, b) that is checked by direct substitution. Hence $F(a, b) = 0$. By Cauchy–Schwarz $|x_{ij}| \leq 1$, so $(a, b) \in [-1, 1]^2$. On the open square F is strictly positive. Indeed the difference $\partial_b F - \partial_a F$ factors as $(a - b)(2a + 2b + 1)$; on the branch $b = a$ the equation $\partial_a F = 0$ becomes $6a^2 + 5a + 1 = (3a + 1)(2a + 1) = 0$, and on the branch $2a + 2b + 1 = 0$ it becomes $-a(2a + 1) = 0$. This gives exactly the four interior critical points $(-\frac{1}{3}, -\frac{1}{3}), (-\frac{1}{2}, -\frac{1}{2}), (0, -\frac{1}{2}), (-\frac{1}{2}, 0)$

with values $\frac{20}{27}, \frac{3}{4}, \frac{3}{4}, \frac{3}{4}$, and the edge restrictions of F are

$$\begin{aligned} F(a, 1) &= 3(a + 1)^2, & F(a, -1) &= 1 - a^2, \\ F(1, b) &= 3(b + 1)^2, & F(-1, b) &= 1 - b^2, \end{aligned}$$

whose only zeros on the closed square are the three corners $(-1, 1)$, $(1, -1)$ and $(-1, -1)$ of the (a, b) -square. An interior zero of F would force an interior critical point with a nonpositive value, and there is none. At the three corners, respectively $x_{24} = 1$, $x_{34} = 1$, and $x_{14} = -1 - a - b = 1$, so two of the d_i coincide by the equality case of Cauchy–Schwarz, contradicting distinctness. No configuration exists. $\square$

Remark 7.7. The certificate identities show that the four linear forms, and with them $F(x_{34}, x_{24})$, lie in the saturation of $\langle g_0, \dots, g_4 \rangle$ by $s_1 s_2$: over the complex numbers, every solution of the distance- $\frac{1}{2}$ system with $s_1 s_2 \neq 0$ is already isosceles-patterned. The certificates were found by a Gröbner-basis saturation in Singular 4.3.2, with the saturation recomputed by iterated ideal quotients under in-run containment and stability checks, and the computed reduced basis of the fully saturated ideal is $\{x_{23} + x_{24} + x_{34} + 1, x_{14} + x_{24} + x_{34} + 1, x_{13} - x_{24}, x_{12} - x_{34}, F(x_{34}, x_{24})\}$; the proof above uses only the certified fragment, re-verified by exact expansion in SymPy 1.14, so no Gröbner implementation is trusted. As a standing positive control, the same pipeline at the distance $\frac{1}{3}$ keeps the regular tetrahedron on the saturated variety, as it must.

Theorem 7.8 (five sites per side). *Two distinct compatible decompositions of one proper-zero raw union differ in at least five reflected sites on each side: $|A \setminus A''| \geq 5$ and $|A'' \setminus A| \geq 5$.*

Proof. As in Theorem 7.2, distinctness makes both difference sets nonempty. By Lemma 7.5 each side of the exchange graph $\mathcal{E}$ has at least four sites, and if some side had exactly four, $\mathcal{E}$ would be $K_{4,4}$ minus a perfect matching with all eight sites at one distance $R > 0$ from o and all twelve edges of length R . Translating o to the origin and scaling by R^{-1} makes the sites unit vectors with $\|a - c\| = 1$, i.e. $a \cdot c = \frac{1}{2}$, along every edge; labelling the matching pairs with equal indices produces exactly the configuration of Theorem 7.6, which does not exist. Hence both sides have at least five sites. $\square$

Corollary 7.9 (at most four hidden facets). *A proper-zero datum with at most four hidden facets has a unique compatible decomposition, with no hypothesis on its centre.*

Proof. A second decomposition would give $|A \setminus A''| \geq 5$, while $|A| \leq 4$. $\square$

The corollary is now also immediate from Theorem 4.6; it is retained because it is the sharp conclusion supplied by the independent exchange route before ridge propagation is invoked.

8. EXACT RECONSTRUCTION ALGORITHM AND EVIDENCE

Assume an exact rational polyhedral presentation from which the exposed polygonal boundary can be computed. Coplanar subdivisions are presentation data and may be inserted or removed. In E-A72 the same boundary is obtained by exact face cancellation from a tetrahedral presentation. The universal conceptual algorithm is:

1. Compute the closures of the connected components of the local-flat locus. With at least three components, select the unique centre of their codimension-one adjacency star; with exactly two, select the component of larger intrinsic volume. This is M_J .
2. Intersect every other component with M_J to recover the nonflat hinges and their cap geometry.
3. Recover o from either nonflat cap by the two-candidate construction in Proposition 3.1.
4. Mark every recovered visible hinge facet. For every edge shared by two already marked facets, record the pairing.
5. Choose an unpaired edge R of a marked facet. In its oriented normal quotient, remove the known boundary ray and retain the other ray, including when the two form one straight line.
6. With $q = \text{proj}_{\text{aff } R}(o)$ and $r = \text{dist}(o, \text{aff } R)$, recover the new site as $a = q + ru$ from that oriented unit ray u .
7. In the strict a -side of the bisector $B(o, a)$, select the connected component of M_J containing a . Its closure is the new cap and its intersection with $B(o, a)$ is the new facet. Mark it and repeat until every facet has been recovered, equivalently until no unpaired frontier edge remains.

Correctness and termination follow from Lemmas 4.3, 4.4, and 4.5, together with connectivity of the facet-adjacency graph, exactly as in Theorem 4.6. E-A72 is an independent exact implementation of this propagation kernel on rational tetrahedral raw presentations. It deliberately does not implement the end-to-end extraction of flat components, nonflat hinges, and the centre in steps 1–3.

There is also an auxiliary one-shot algorithm: construct $\Gamma(M_J)$, keep its passing vertices, and intersect the visible hinge halfspaces with their bisector halfspaces, as in the Dirichlet-cell construction of [3]. It is correct under Theorem 5.6 and is useful as an independent implementation check, but it is not the basis of the universal theorem. A full certifier may rebuild the entire raw union, including the observed nonflat caps, and compare it exactly with the input. The reference signature code implements this flat reconstruction kernel, not steps 1–3 or a full raw-union certifier.

The certifying variant compares two intrinsic conditions, and each is provably necessary. There are constructed exact data, at exceptional centres, on which the reconstruction returns a strict superset of the true site set while the rebuilt merged flat component still equals the observed one exactly:

cutting a passing vertex off with its bisector plane and re-attaching the cap over the new facet restores the removed piece. Comparing only the merged flat component therefore certifies a wrong answer on those data (six exact witnesses). What the wrong reconstruction destroys is the recovered visible hinge facets, so the certifier also requires every recovered visible hinge facet to survive as a facet of the reconstructed core. On a second witness family (three exact witnesses) the situation is reversed: the hinge facets survive while the flat components differ. Hence neither condition alone suffices, and the certifier checks both.

For the signature reference implementation, let N be the number of input boundary corners and k the number of returned core facets. Its germ extractor atomises each segment against all exact endpoints and builds carrier labels during atomisation. Its conservative bound is $O(N^2 \log N)$ exact field operations and $O(N^2)$ crease atoms. The reference three-dimensional halfspace kernel enumerates the $\binom{k}{3}$ plane triples and validates each candidate corner against every halfspace, so it costs $O(k^4)$. Thus the proved reference bound is

$$O(N^2 \log N + k^4).$$

No sharper literature bound is imported into this statement. No corresponding worst-case bound is claimed here for the independent propagation engine.

The executable audit has two signature layers and two propagation layers. E-A60 runs the intrinsic germ skeleton on every frozen six-vertex row, and E-A58 runs the reconstruction/flat certification kernel. E-A71 checks the oriented local ridge lemma; E-A72 independently implements full facet-graph propagation with site-anchored component selection. The frozen measurements are:

intrinsic-skeleton corpus rows	16,744
rows reconstructed by the intrinsic signature	16,744
rows agreeing with the historical graph on the corpus	16,744
subdivision mutations invariant (256-row corpus prefix)	256/256
maximal-sheet cross-checks (same prefix)	256/256
flat-kernel reconstructions	16,744/16,744
sampled two-condition certifications	340/340
wrong answers certified	0
degenerate swallowed-corner witness	pass
exceptional-centre witnesses	6
flat-only accepts / two-condition rejects	6/6
reversed witnesses (hinges survive, flat differs)	3/3
corpus disagreements between the two certifier variants	0/340
certificate identities of Theorem 7.6 verified	4/4
oriented local data (E-A71)	16,759
oriented local ridge steps (E-A71)	189,128
global propagation data (E-A72)	16,759
global BFS/component steps (E-A72)	61,030
visible-hidden / hidden-hidden (E-A72)	35,228 / 25,802
straight coplanar fusions (E-A72)	445
ridge-touch cases, site-disambiguated (E-A72)	34,090
global propagation mismatches	0
propagation mutation kills	4/4

The canonical payloads are, for the intrinsic-skeleton audit E-A60,
D18C07176BAF7B729A550E9B9B8B9D9E04BB58A0C52D295243108417C9BE785A,
for the two-condition reconstruction/certification kernel E-A58,
3C58C0B88081380C7D22993A6FEE3B4EC9FE1A8C3D0CDC107BE9D40C33BA4ACC,
for the exceptional-centre adversarial control E-A61,
76956805B0BA304142CB08D833FA1179876A9F2154A76CE6C5A232BEF9E3F7F1,
for the reversed witness family E-A63,
50B26633575C81C67AD3A9857DC5BE1227B512E21CC98168EECEC37B360DB1A8,
and for the certificate verification of Theorem 7.6, whose payload pins the
SHA-256 of all four cofactor files, E-A69,
5DC4C5800572C64F07C7A815CE7BEFFB930FEF03E04CC71679FB9501606F1D92.

The oriented local-ridge audit E-A71 has payload

3EA0CA24EB09B31099F9503B0DCF84C1D7D02EB9894C73F0C299E08CF4732BC3,
and the independent global propagation audit E-A72 has payload

46F44A8E7444B395AF987C2BA5EA9A0BF1DFBD69E6BA782525FA71822D42D4AA.

Artifact availability. The exact source, scripts, frozen result records, certificate files, tests, and verification instructions underlying the cited E-A58, E-A60, E-A61, E-A63, E-A69, E-A71, and E-A72 audits are bundled in the companion repository <https://github.com/aconsciousfractal/>

FCIG-Generic-Reconstruction-from-Erased-Flat-Boundaries. The E-A identifiers are local experiment locators, and the displayed SHA-256 values bind their canonical contents; they are not literature citations.

Finite replay is evidence about the implementation and its frozen domain. The proof of Theorem 4.6 is the intrinsic argument in the preceding ridge-propagation section. The signature audit separately supports Theorem 5.6, and the proof of Theorem 7.6 rests on the verified identities and the elementary analysis of Section 7, not on replay.

9. RELATION TO SUPPLIED-BOUNDARY INVERSE PROBLEMS

Ash and Bolker study recognition of supplied Dirichlet tessellations. Their Lemma 1 places a common cell boundary on the perpendicular bisector of its two sources, and their Theorem 2 constructs a Dirichlet cell by reflecting an interior point in the known facet hyperplanes [3]. The latter is exactly the forward reflection construction used in Section 2 and is cited as such.

Subsequent recognition algorithms also begin with substantially visible boundary data. Aurenhammer takes a supplied polytopal cell complex [4]; Hartvigsen takes a tessellation of Euclidean space into polyhedra and uses linear programming [15]; and Schoenberg, Ferguson, and Li recover generators from segments demarcating cell boundaries [18]. In the plane, Alonso Ferrero states the generator-recognition problem with the Voronoi edges provided [13]. Biedl, Held, and Huber reconstruct from a supplied embedded straight-line graph with rays [5]. Power-diagram detection and weighted bisector-graph reconstruction similarly start from a supplied complex, partition, or geometric graph [7, 12]. The generalized inverse Voronoi problem studied by Aloupis et al. permits added edges while retaining the original input edges [2].

Several closer inversion routes still assume that the tessellation itself is supplied. Suzuki and Iri fit a planar Voronoi diagram to a given planar tessellation; their formulation includes generator restoration when that input already is a Voronoi diagram [19]. Duan et al. recover weighted generators from a complete Laguerre tessellation and show that this inverse problem is generally nonunique [11]. On the sphere, Chaidee and Sugihara recognize a supplied spherical Laguerre tessellation via a centrally projected polyhedron, again with nonunique generator sets [10]. Most closely in propagation style, Hernandez-Suarez begins with a complete planar Voronoi tessellation, solves for one interior cell and its neighbours, and obtains the remaining sites by successive geometric reflections [16]. In every case the cell boundaries, partition, or tessellation adjacency remain available; they have not been erased into the unlabelled raw union considered here.

Not every inverse tessellation problem receives boundary edges. Bourne, Pearce, and Roper prove that labelled Laguerre cells are uniquely determined by their volumes and centroids and recover them by convex optimization [9]. Bourne, Mulholland, Sahu, and Tant instead fit an oriented planar Voronoi

diagram from ultrasonic boundary travel times in a simplified nondestructive-testing model; the geometry is unknown, their first objective can have nonunique minimizers, and the reported reconstruction is noise-sensitive [8]. These are partial- and indirect-data comparators, but their observables are respectively global labelled cell moments and travel times, not a reflected-cap raw union.

Three further classical theories are sometimes suggested as antecedents. They concern different observables: the theory of convex polyhedra from intrinsic surface data [1], the reconstruction of a simple polytope from its graph [6, 17], and geometric tomography, which works from lower-dimensional data such as projections and sections [14]. None of these takes the reflected-cap raw union as its input, and no comparison beyond that input-model distinction is made here.

The precise distinction in the present model is that the zero-angle source layer and its facet boundaries have already been erased by the raw union. The intrinsic germ skeleton is designed to use only the boundary information that survives that erasure. This comparison is a statement about input hypotheses, not a claim that the present result is the first possible treatment of every related erased-boundary problem.

10. LIMITATIONS AND OPEN PROBLEMS

The following questions are not resolved here.

- All-zero data lie outside the theorem: with no nonflat cap there is no visible hinge from which to recover the centre and start propagation. Whether a three-dimensional all-zero raw union can conceal two noncongruent structural histories remains open.
- The proof is three-dimensional. The normal-section calculation itself is dimension-independent, but intrinsic initialization and global recovery have not been promoted beyond dimension three.
- Theorem 7.6 is computer-assisted: the four certificate identities were found by machine and are re-verified by exact expansion. The verification chain ships with the reference implementation; the remaining steps are hand-checkable.
- The reference implementation assumes an exact polygonal presentation. It does not extract such a presentation from a point cloud, floating-point mesh, or arbitrary implicit-set oracle.
- E-A72 implements the exact propagation kernel on a tetrahedral corpus; it is not the end-to-end extraction in steps 1–3 or a general polyhedral component engine. No numerical-stability or noise theorem is claimed.
- E-A58 implements the auxiliary signature reconstruction kernel and its two-condition certifier; it is not the full pipeline or a raw-union certifier. Completeness of the two conditions is not claimed; each

is necessary, and their conjunction rejects every constructed wrong reconstruction.

- Neither the finite corpus, the deterministic mutation sample, nor the mathematical proof establishes novelty or priority. The finite evidence also does not establish unrestricted-domain behaviour.

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